Designing an Algorithmic System for Energy Market Forecasting

Benchmark EPF models inside an **MLOps platform**

Bachelor's thesis defense

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DTU Compute · Supervisor: Finn Årup Nielsen · June 2026

Forecasting models must be operated, not just built

EPF research

- Produces accurate benchmark models.
- Evaluates them **offline, once**.

Production reality

- A served model must be **retrained, monitored, reproduced, audited**.
- That operational layer is what **MLOps** addresses — and EPF research leaves out.

This thesis

Build a production **MLOps platform** that operates benchmark EPF models, and **evaluate the platform itself**. The two fields rarely meet (only Subramanya et al.).

Aim and research questions

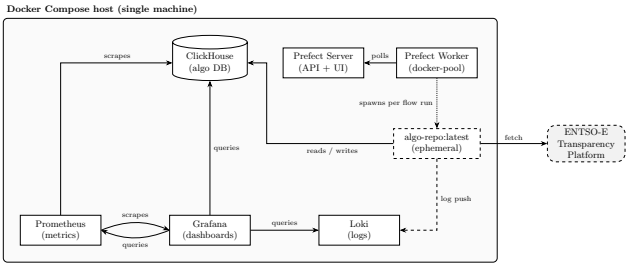
A platform that ingests ENTSO-E data, recalibrates and forecasts **daily** for DK1 and DE, **monitors** quality and drift, and keeps **full provenance**.

RQ1 **Forecasting** — how do replicated LEAR, DNN and Transformer compare? *(the workload the platform serves)*

RQ2 **Engineering** — what trade-offs emerge when productionising these models?

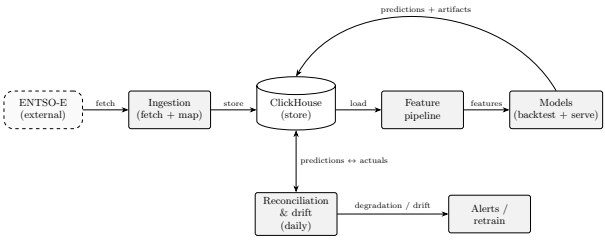
RQ3 **Maturity** — which of Zarour et al.'s nine MLOps practices does the platform realise, and where does a single-host MVP reach its limit?

The platform (§4.2)



- One Docker host, **six services**: ClickHouse, Prefect (server + worker), Prometheus, Grafana, Loki.
- Flow runs spawn ephemeral `algo-repo` containers.
- Daily loop (UTC): **06:00** forecast, **18:00** reconcile, **19:00** drift.
- Serves three benchmark EPF models on DK1 and DE.

One data path, end to end (§4.2)



- ENTSO-E → XML mappers → ClickHouse (`timeseries`, `forecasts`; `ReplacingMergeTree`).
- Preprocessor → backtest / live runner → predictions + artefacts.
- Reconciliation back-fills actuals; drift compares distributions and can raise alerts or retraining.

Built around nine MLOps practices (Zarour et al.)

P1 CI/CD	GitHub Actions	**P6** Train & evaluate	daily forecast + reconcile
P2 Orchestration	Prefect flows	**P7** Metadata	ClickHouse + W&B
P3 Reproducibility	GIT_SHA + Compose	**P8** Monitoring	Grafana / Loki / Prometheus
P4 Versioning	Git + migrations	**P9** Feedback loops	drift → retrain (opt-in)
P5 Collaboration	PRs + versioned docs		

These structure the system chapter — and I will show seven of them running **live, next**.

▶ LIVE DEMO

Adding a new market (DK2), end to end

CI/CD · Prefect · Grafana · W&B

“A contributor adds DK2 and ships it to a live, monitored forecast.”

seven of the nine practices — running

The workload: three benchmark models (§3.1, 3.3)

- **LEAR** — one Lasso per delivery hour (Lago et al.).
- **DNN** — MLP mapping the feature vector to all 24 hours at once (MIMO).
- **Transformer** — encoder-only, two input branches (Llorente & Portela).

Replication framing

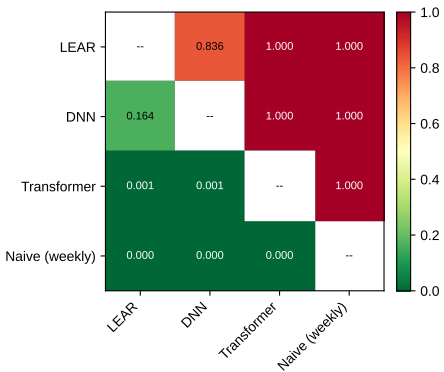
Faithful **in specification**, on current DK1 + DE data. Because the price regime differs from the 2016–2018 source-paper era, the comparison is read in **relative** terms (rMAE and rankings), with paired **Giacomini–White** significance tests.

RQ1 — accuracy and significance (§5.1–5.2)

rMAE (common days, ↓ better)

	DK1	DE
LEAR	**0.469**	**0.376**
DNN	0.482	0.388
Transformer	0.543	0.420
Naive (weekly)	1.000	1.000

- LEAR ≈ DNN, **indistinguishable** ($p = 0.16/0.19$).
- Both **beat** the Transformer ($p \leq 0.013$); all beat the naive.



Giacomini–White p-values, DK1.

RQ1 — cost, and ranking vs the papers (§5.3–5.4)

Cost (s / recalibration-day)

	DK1	DE
LEAR (CPU)	50.3	41.3
DNN (GPU)	0.9	1.6
Transf.(GPU)	1.6	1.9

The DNN gives **equal accuracy for far less compute**.

- Our order LEAR $\geq$ DNN $\geq$ Transformer **inverts both papers**.
- LEAR–DNN inversion is only numerical (they are tied).
- Transformer's third place is genuine and significant.
- **Most consistent with** the deep models' reduced tuning budget (DNN 30 trials vs the paper's 1500), not weaker architectures — a hypothesis, confirmatory run is future work.

RQ3 — maturity and quantitative evaluation (§6.1–6.4)

Practice	Status	Lvl
P1 CI/CD	Realised	2
P2 Orchestration	Realised	2
P3 Reproducibility	Realised	2
P4 Versioning	Realised	2
P5 Collaboration	Realised	1
P6 Train & evaluate	Realised	1
P7 Metadata	Realised	1
P8 Monitoring	Realised	2
P9 Feedback loops	**Minimal**	1

8 of 9 realised; 1 minimal. Backed by numbers:

- **ML Test Score 2.5** (min of sections), bound by data testing.
- Provenance **66.7%**, 100% from the first CD run.
- A logged forecast re-ran to 7×10^{-6} **EUR/MWh**.

DevOps practices transferred easily; ML-specific data testing is where the MVP is weakest.

RQ2 — engineering trade-offs (§7.2)

- Productionising the benchmark models **mostly aligned** the needs of a faithful replication and a single-host MVP — both favour determinism and a single source of truth.
- **Full refit from scratch** each day: cheap, avoids warm-start error, makes drift-triggered retraining redundant.
- **One OLAP store** for data, predictions, metrics, artefacts ⇒ reconciliation is one query.

A genuine tension I resolved

The Transformer's native *train-once* + *renormalise* scheme vs the daily-refit pipeline. I chose production consistency and deviated from the paper.

Conclusion and future work

Contribution

A reproducible, monitored MLOps platform that operates three benchmark-faithful EPF models, with a quantified **maturity evaluation** and a significance-tested model comparison — joining MLOps and EPF in one artifact.

Result: 8 of 9 practices realised; ML Test Score 2.5; reproducible to 7×10^{-6} EUR/MWh. LEAR and the DNN tied at the top, Transformer weaker.

Future work: probabilistic forecasting (CRPS); close the replication gap (full tuning + `epftoolbox` side-by-side); recalibrate drift and broaden coverage; controlled re-run of the stochastic models.

Thank you — questions?

Backup slides follow: the nine practices in detail, system internals, and forecasting / statistics deep-dives.

Backup slides

implementation detail · the nine practices · forecasting & statistics

Backup — tech stack and the alternatives rejected

Role	Chosen	Over
Storage / OLAP	ClickHouse (one store)	a filesystem model store
Orchestration	Prefect	Airflow, Dagster
Runtime	Docker Compose	Kubernetes
Logs / metrics / dash	Loki + Prometheus + Grafana	the ELK stack
Experiment tracking	Weights & Biases	MLflow (deferred)
CI/CD	GitHub Actions	Jenkins, CircleCI
Secure deploy access	Tailscale (ephemeral SSH)	a public SSH port

Every choice sized to a **single-host MVP**.

Backup — P1 Automated CI/CD (§4.3.1)



- PR gates: `pr_test`, `pr_ruff-lint`, `pr_smoke`.
- On merge: `cd_deploy` over Tailscale, migrations, image rebuilt with the commit SHA, Prefect redeployed, smoke forecast.

Backup — P2 Orchestration (§4.3.2)

- Four Prefect flows on a Docker work pool: ingestion, forecast, reconciliation, drift.
- **16 deployments** from shared definitions, parameterised by area (one definition, one deployment per zone).
- One task per data type, sequential, `retries=2`; fetch/insert logic in a shared service with no Prefect dependency.

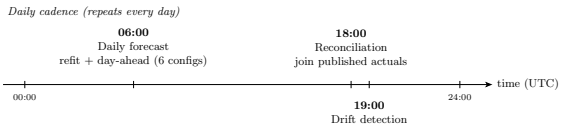
Backup — P3 Reproducibility (§4.3.3)

- Whole stack in `docker-compose.yml`; rebuildable from the repo alone.
- `GIT_SHA` baked into the image; local runs emit empty so a dirty tree never stamps a misleading SHA.
- Config-driven pipeline: calibration windows, `asinh` transform, enumerated seeds all pinned.

Backup — P4 / P5 Versioning and collaboration (§4.3.4–4.3.5)

- Git tracks `configs/`, numbered SQL migrations, Grafana dashboards and alert rules.
- DVC evaluated and rejected — ENTSO-E data is replayable; `ReplacingMergeTree` + `git_sha` give provenance without a binary store.
- Every change via PR; conventions in versioned `REPO_OVERVIEW` / `RUNBOOK` / `CONTRIBUTING`.

Backup — P6 Continuous training and evaluation (§4.3.6)



- Six `daily-forecast` deployments refit at 06:00; the DNN averages a 4-seed ensemble.
- Each run refits from scratch on a rolling window.
- Reconciliation scores live forecasts next day with the same metrics as the backtest.

Backup — P7 Tracking ML metadata (§4.3.7)

- ClickHouse: `experiment_runs` (git SHA, data window, duration), `experiment_predictions`, `model_artifacts`, `hyperparam_trials`.
- W&B research surface: `algo-research` (backtests, sweeps), `algo-prod` (reconciliation). `WANDB_MODE=disabled` no-ops it in CI.

Backup — P8 Continuous monitoring (§4.3.8)

- Three Grafana dashboards: Data Quality, System Health, Operational (Loki stream).
- Prometheus scrapes ClickHouse / Grafana / itself every 15 s.
- Five alert rules: staleness, drift-detected, reconciliation-missed, ClickHouse-down, accuracy-degradation ($> 1.5\times$ 30-day median).

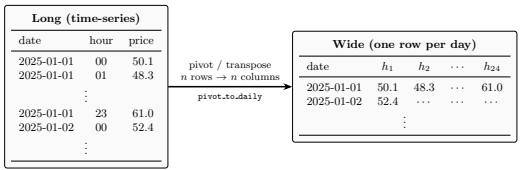
Backup — P9 Feedback loops (§4.3.9)

- Drift (Evidently, 90-day vs 7-day) daily in-process; results in `drift_reports`; detected events alert and can trigger an audited retrain.
- Trigger **off by design** — the cheap daily refit keeps models fresh (cost-aware argument of Mahadevan & Mathioudakis).
- The one **minimal** practice. Drift flagged 49/49 (over-powered) — next step is seasonally-matched windows and testing residuals.

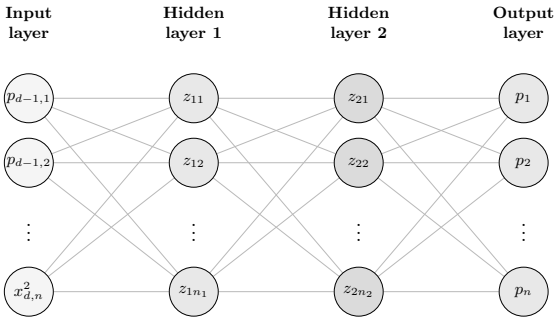
Backup — features and the target transform (§3.2)

$$\underbrace{4n}_{\text{price lags}} + \underbrace{2 \cdot 3n}_{\text{2 exog} \times \text{3 lags}} + \underbrace{7}_{\text{weekday}} = 247$$

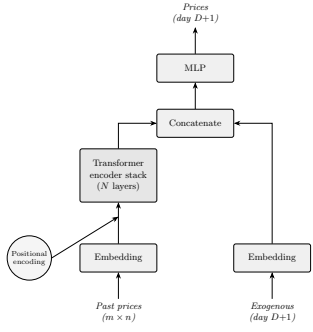
($n = 24$). LEAR's Lasso shrinks most coefficients to zero; `asinh` tames spikes while staying defined for **negative** prices.



Backup — the two deep architectures (§3.3.2–3.3.3)

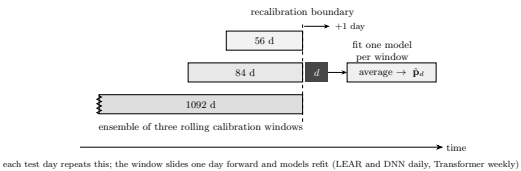


DNN (MIMO MLP)



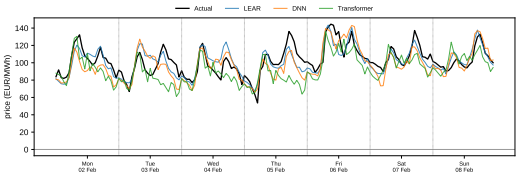
Transformer

Backup — recalibration, evaluation, tuning (§3.4–3.7)

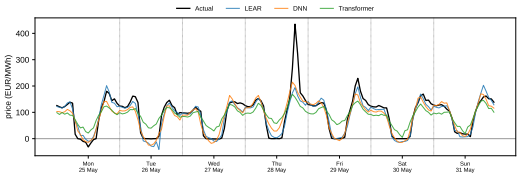


- Rolling walk-forward; forecasts averaged over a calibration-window ensemble; daily refit (LEAR, DNN), weekly (Transformer).
- Metrics MAE / RMSE / sMAPE / rMAE; paired GW tests.
- DNN tuned by Optuna TPE (6 HPs) on an ex-ante validation window.

Backup — why the Transformer is third



Calm week



Renewable week

It smooths over the negative hours and the scarcity peak; LEAR and the DNN follow them.

Backup — Giacomini–White vs Diebold–Mariano

GW regresses the daily loss differential on an instrument set:

$$\Delta_d^{A,B} = \phi' X_{d-1} + \varepsilon_d.$$

- X_{d-1} = constant *and* lagged differentials $\rightarrow$ conditional predictive ability.
- X_{d-1} = **constant only** (Lago eq. 15) $\rightarrow$ multivariate **DM**, which is what I compute, with Newey–West HAC errors (lag $\lfloor T^{1/4} \rfloor$). I follow Lago's naming; I do not exercise the conditioning.

Backup — ”Tied” means indistinguishable, not equal

- $p = 0.16$ is a **non-rejection**, not proof of equality.
- $n \approx 187$ daily differentials; the gap is $\sim$3% relative — power is low.
- Equality would need a TOST / equivalence test; I claim **indistinguishability**.
- No multiplicity correction, but the load-bearing results clear any correction by orders of magnitude.

Backup — documented deviations from the papers (Table 3.4)

Aspect	Paper	This work (reason)
Test length	$\geq$1–2 years	31 weeks (data history)
DNN tuning	1500 TPE iters	30 trials (compute)
DNN search space	arch + feature selectors	architecture only
LEAR windows	56/84/1092/1456 d	56/84/1092 d (1456 predates data)
DNN window	4 years	730 d
λ select	LARS+AIC	`LassoCV` (close, standard)
Resolution	hourly	15-min $\rightarrow$ hourly mean
Transformer regime	train-once + renorm	rolling window + weekly refit
Transformer HPs	tuned per dataset	paper defaults reused